

21 . Spirally traversing a matrix

```
class Solution {
public:
    vector<int> spirallyTraverse(vector<vector<int>> &mat) {
        vector<int> ans;
        int n = mat.size();
        if (n == 0) return ans;
        int m = mat[0].size();
        int top = 0, bottom = n - 1;
        int left = 0, right = m - 1;
        while (top <= bottom && left <= right) {
            // 1) left -> right along top row
            for (int j = left; j <= right; ++j) ans.push_back(mat[top][j]);
            ++top; // 2) top -> bottom along right column
            for (int i = top; i <= bottom && left <= right; ++i) ans.push_back(mat[i][right]);
            --right; // 3) right -> left along bottom row (if still valid)
            if (top <= bottom) {
                for (int j = right; j >= left; --j) ans.push_back(mat[bottom][j]);
                --bottom;
            } // 4) bottom -> top along left column (if still valid)
            if (left <= right) {
                for (int i = bottom; i >= top; --i) ans.push_back(mat[i][left]);
                ++left;
            }
        }
        return ans;
    }
}
```

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Compilation Results' and 'Problem Solved Successfully' with a green checkmark. Below this, statistics are listed: 'Test Cases Passed: 1115 / 1115', 'Attempts: Correct / Total: 1 / 1', 'Accuracy: 100%', 'Points Scored: 4 / 4', and 'Time Taken: 0.29'. At the bottom left, there are buttons for 'Solve Next' and 'Stay Ahead With:' followed by links to 'Find kth element of spiral matrix', 'Rotate by 90 degree', and 'Reverse Spiral Form of Matrix'. The main area on the right shows a C++ code editor with a solution for the 'Spirally Traversing a Matrix' problem. The code is a class 'Solution' with a public method 'spirallyTraverse' that takes a vector of vectors of integers and returns a vector of integers. The code implements the spiral traversal logic using four loops for top row, right column, bottom row, and left column, updating boundaries (top, bottom, left, right) as it goes. The code is numbered from 1 to 35.